CEN263 Digital Design Make-up Exam
Feb 12, 2021

Question:	1	2	3	Total
Points:	30	30	40	100
Score:				

- Read all directions carefully.
- This exam must be completed individually. **Cheating is not allowed and may result in you receiving a mark of zero for the module affected.**
- You should write your answers on paper. You can use more than one paper if you need it. Please do not forget to write your name and student id every paper you used.
- You should scan your papers into a single .pdf file before submission. It is recommended that you use an App on your phone for ease of use with on-device camera for creating electronic submissions. Select an App that is able to convert images into .pdf files. It is beneficial to have the functionality to detect the edges, correct the orientation, and remove/cleanup the background.
- You must show all your work to receive full credit. Messy or unreadable submissions will receive no credit.
- You have 40 minutes to complete this exam. Good luck.

1. (30 points) Simplify the following function and implement it by using two-level NAND gates.
F(A,B,C,D) = A'B'C +AC' +ACD + ACD' + A'B'D'

2. (30 points) Design a combinational circuit that accepts a three bit number and generates an output binary number equal to the square of the input.

3. (40 points) Implement a full adder circuit by using:
a) 3 – to - 8 line Decoder (20 pts)
b) 4 X 1 Multiplexers (20 pts)